



Biology H Review Practice Answer Key

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Range: Chapter 12-13

Note: (Review Guide/Review Practices/Practice Set) are UNOFFICIAL sources of reference and do NOT include o hint at questions in the actual exam!

I. MCQ

1. B 2. C 3. A 4. B 5. B 6. B 7. A 8. B 9. C 10. C 11. B 12. B 13. B 14. B 15. B
16. A 17. C 18. C 19. B 20. B

II. FRQ

1.

(a)

Prokaryotic DNA: Single circular chromosome in cytoplasm; no telomeres.

Eukaryotic DNA: Multiple linear chromosomes in nucleus; telomeres protect ends from shortening.

(b)

Okazaki fragments: DNA polymerase synthesizes only 5'→3', but lagging strand runs from 3' to 5'. Lagging strand is replicated away from the replication fork → must be synthesized as short Okazaki fragments.

Enzyme: DNA polymerase I replaces primers with DNA sequence; DNA ligase joins fragments

(c)

Increased mutations due to uncorrected errors, leading to diseases like cancer

2.

(a)

mRNA: 5'-AUG CAU GAA CGU-3'

Protein: Methionine (AUG) - Histidine (CAU) - Glutamic Acid (GAA) - Arginine (CGU)

(b)

Mutation type: Substitution.

Amino acid change: Original: Arginine (CGU) → Mutated: Proline (CCA).

(c)

5' cap/3' poly-A tail: Protect mRNA from degradation; enable ribosome binding.

Splicing: Removes introns (non-coding) → joins exons for the production of more functional protein from one DNA sequence.

3.

(a)

Cancer cells: Telomerase highly active → prevents telomere shortening → unlimited division

Somatic cells: Telomerase inactive → telomeres shorten → limits cell division (except for stem cells or fast-growing cells)

(b)

Plants: Beneficial → larger/stronger

Animals: Lethal